Navigating AI in Higher Education

Student Views on Professors' Attitude of AI Tools in the Classroom Analyzed through Sentiment Analysis and Topic Modeling

Sentiment Analysis · Topic Modeling · Reddit Data · Artificial Intelligence

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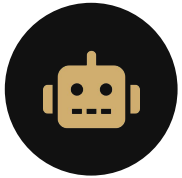
CONTEXT

AI Is Reshaping Higher Education



Faculty & Student

Shared strong interest in AI but a lack of clear instructions and standardized guidance for implementing it effectively in academic settings



Development of AI

Increasing integration of technology into learning environments, research and practice related to the use of artificial intelligence (AI) in education have expanded rapidly.



Social Media

Discussions around AI and technology have surged on social media platforms

(openici & Kerr, 2017; Shanmugasundaram & Tamilarasu, 2023; Waterloo et al., 2017 ;MacDonald et al., 2024; Wu et al., 2024; Yan & Liu, 2022)

THE GAP

What's Missing?

What We Know

- Institutional AI policies
- Faculty‘s attitudes toward AI
- Student‘s perspective toward AI

What We Don't Know

How do students perceive their professors' attitudes toward AI?

Students interpret faculty attitudes toward AI through course materials, academic integrity decisions, and peer communication rather than through open discussions.

RESEARCH QUESTIONS

01 What sentiment trends (positive, neutral, negative) emerge in posts discussing faculty attitudes toward AI?

02 What topics or concerns do students express when referring to professors' attitudes toward generative AI?

DATA

Listening to Students on Reddit

Anonymous platform enables candid, unfiltered, direct student expression

6 Subreddits

- r/AskProfessors
- r/AIEducation
- r/Student
- r/CollegeRant
- r/University
- r/college

3,621 posts collected

1,001 after filtering

Jan 2023 – May 2025

Keywords: AI, ChatGPT, GPT-4, generative AI, LLM, Claude, classroom …

METHOD

Analysis Pipeline

01



Collect

Reddit posts
via PRAW API

02



Filter

Fuzzy Matching

03



Sentiment

RoBERTa model
pos / neu / neg

04



Topics

BERTopic for
theme discovery

Both methods leverage pre-trained language models — no manual coding required.

METHOD

Data filter process

01



Collect

Reddit posts
via PRAW API

Identify opinions about attitude towards AI usage within teaching and learning process

02



Filter

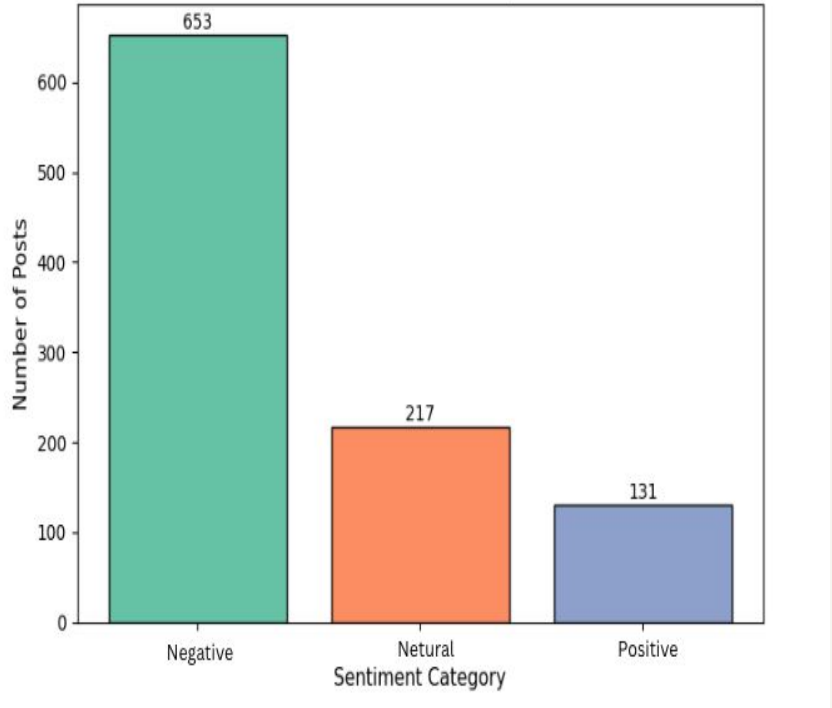
Fuzzy Matching

Keyword matching to identify students: rule-based keyword matching strategy: any post containing three or more of these keywords was classified as **representing a student perspective.**

Both methods leverage pre-trained language models — no manual coding required.

FINDINGS · RQ1

Sentiment Distribution



65% **Negative**

Reflect negative perceptions of professors' attitudes toward the use of generative AI in education

22% **Neutral**

Mixed feelings

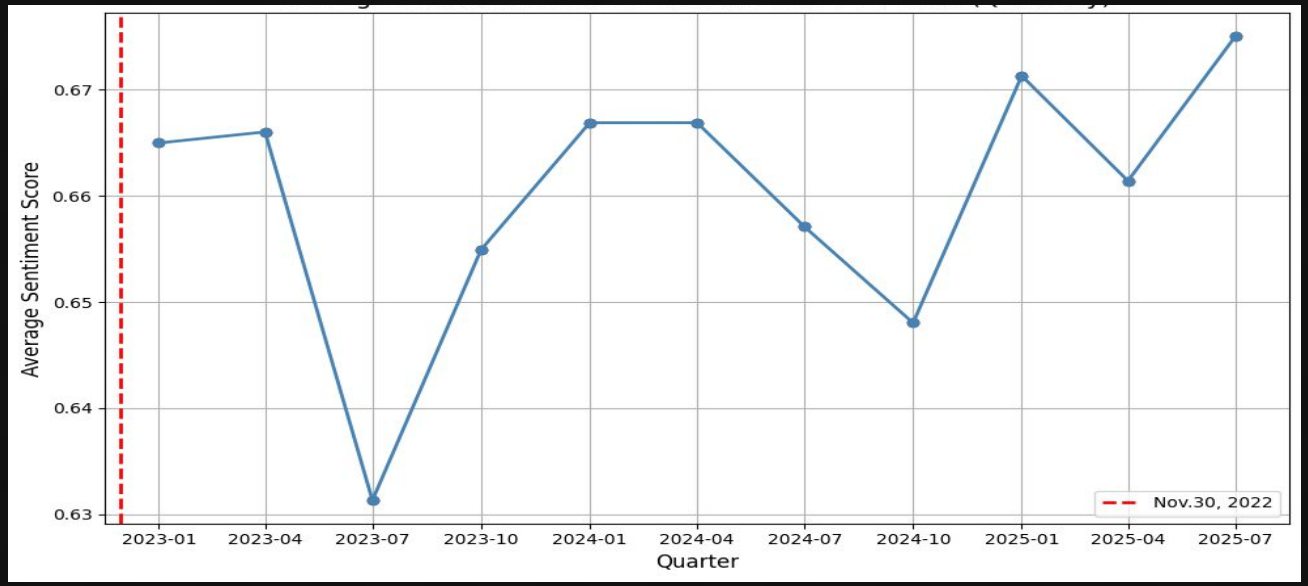
13% **Positive**

Feel supportive from they professors' attitude towards AI

FINDINGS · RQ1

Sentiment Over Time

Red line: ChatGPT release

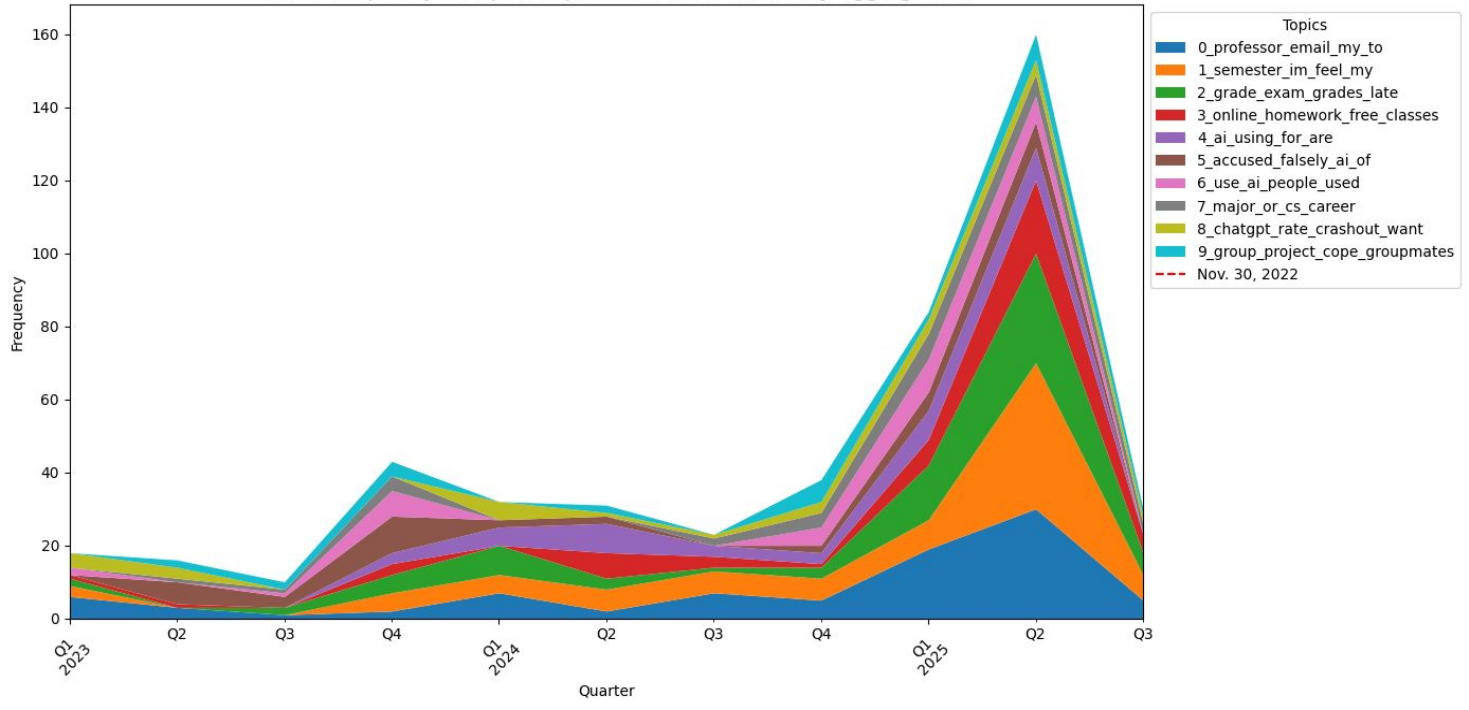


Lowest: Q3 2023 (0.631)
Peak uncertainty after ChatGPT launch

Recovery from Q4 2023 onward
Steady improvement into 2025

FINDINGS · RQ2

Concerns Are Intensifying



Notable surge in 2025

As GenAI becomes more embedded in academics, students increasingly seek clarity on faculty attitudes.

FINDINGS · RQ2

What Are Students Talking About?

Communication Gaps

Delayed emails, unclear expectations from faculty

Academic Integrity

Fear of false AI accusations, unclear rules

Grading Concerns

Exam delays, fairness in AI-era assessment

Emotional Stress

Academic workload, anxiety about AI policies

Remote Learning & Access

Course accessibility, tech requirements

Shaped by institutional clarity and supportive communication

IMPLICATIONS

What Does This Mean for Practice?



Open the Dialogue

Faculty should proactively communicate AI policies rather than relying on syllabi or disciplinary actions alone



Clear Institutional Guidance

Universities need consistent, transparent AI use policies that address student concerns about fairness and integrity



Leverage Student Voice

Social media data reveals authentic student perspectives that can inform policy development and faculty training

Limitations

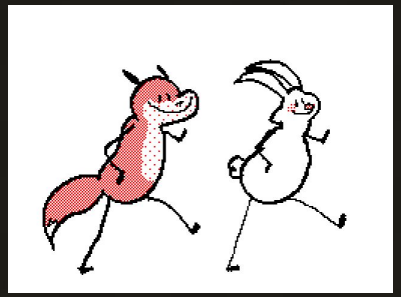
- Reddit users may not represent all students (e.g., international students)
- Keyword-based filtering may miss nuanced student expressions

Future Directions

- Interviews, discourse analysis
- Expand to additional platforms beyond Reddit
- Implement more sophisticated student-identification filters

Thank You

Questions & Discussion



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